

LIST OF PROBLEMS: AUCTION WITH COMPLETE INFORMATION

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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1 Brief Math preliminaries

1.1 Consider the function

$$f(x | v, p) = (v - p) \cdot \mathbf{1}_{\{x > p\}}(x), \quad x \in \mathbb{R},$$

where $v, p \in \mathbb{R}$ are arbitrary real parameters and $\mathbf{1}_{\{x > p\}}(x)$ is an *indicator function*:

$$\mathbf{1}_{\{x > p\}}(x) = \begin{cases} 1 & \text{if } x > p, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Solve, for all parameters $v, p \in \mathbb{R}$, the maximization problem:

$$\max_{x \in \mathbb{R}} f(x | v, p).$$

This means that you should find, for all $v, p \in \mathbb{R}$, the set of values of x that maximize $f(x | v, p)$, i.e.,

$$X^*(v, p) = \arg \max_{x \in \mathbb{R}} f(x | v, p).$$

(b) Is it always the case that $v \in X^*(v, p)$?

2 Standard Auction Formats

For the auctions in this section, assume that the tie-breaking rule is the one I discussed in class (in the case of a tie, the winner is the player with the lowest index) unless I tell you otherwise.

2.1 Find all the Nash equilibria of a SPA with two bidders. You are encouraged to do this by constructing and graphing the players' best response functions.

2.2 Show that in a Nash equilibrium of the FPA the two highest bids are the same, one of these bids is submitted by player 1, and the highest bid is at least v_2 and at most v_1 . Show also that any action profile satisfying these conditions is a Nash equilibrium.

2.3 Consider the FPSA (first price sealed-bid auction) with complete information that we discussed in class, and suppose there are two bidders. Assume that bids and valuations are restricted to be multiples of some discrete monetary unit ϵ (e.g. a tiyn). The tie-breaking rule is that if the two bidders submit the same bid, each gets the object with probability $1/2$.

(a) Give a complete description of the best response function of player i as a function of the bid b_j of the other player.

(b) Using the description of the best response function, answer the following questions:

- i. Are any strategies strictly dominated?
- ii. Are any strategies weakly dominated?
- iii. Is there a Nash Equilibrium? What is it? Is it unique?

2.4 Consider the variants of the FPA and SPA in which *all bidders* pay the winning price. Suppose there are two bidders.

(a) Model these two variants as a game by writing down the payoff functions for each player under the two variants.

(b) Does this variant of the SPA has an equilibrium in pure strategies? What is it?

(c) Show that this variant of the FPA has no equilibrium in pure strategies.

2.5 Consider the standard all-pay auction we considered in class, with each bidder having a common value $V > 0$ for an object which is perfectly divisible, i.e., a uniform tie-breaking rule. Suppose that there are $n \geq 2$ bidders. Find a mixed strategy Nash equilibrium in which each player uses the same mixed strategy.

3 Extensions

3.1 Consider a *third-price* sealed-bid auction which differs from the FPA and SPA only in that the winner, which is the player who submit the highest bid, pays the third highest bid. Assume that there are at least three bidders.

- Show that for any player i , the bid of $b_i = v_i$ weakly dominates any lower bid but does not weakly dominate any higher bid.
- Show that the action profile in which each player bids her valuation is not a Nash Equilibrium.
- Find at least two Nash equilibria.

3.2 Two players have a common valuation for an object given by $V > 0$. Suppose that they submit sealed bids $b_i \geq 0$, for $i = 1, 2$. However, the *probability* of winning the object by player i is now determined by the function $p_i(b_i, b_j)$ given by

$$p_i(b_i, b_j) = \begin{cases} \frac{b_i}{b_i + b_j} & b_i > 0, b_j > 0 \\ 1 & b_i > 0, b_j = 0 \\ 0 & b_i = 0, b_j > 0 \\ \frac{1}{2} & b_i = b_j = 0. \end{cases}$$

If player i submits b_i it becomes a sunk cost, i.e., she needs to pay her bid. Thus, player's i payoff is given by

$$u_i(b) = p_i(b_i, b_j) \cdot V - b_i.$$

In this exercise you will find the symmetric Nash Equilibrium of this game.

- Start by maximizing player i 's payoff, assuming an *interior solution* for b_i , i.e., $b_i > 0$, and assuming that $b_j > 0$. Compute the best response functions for both players when both bids are positive.
 - Use the best responses to find the *unique symmetric equilibrium* of the game.
 - Can you generalize this exercise to n players?
- 3.3 Two players compete for an indivisible object whose value is 1 for each player. In each period, each player can either give up or fight. The game ends when a player gives up, and in this case the object is obtained by the other player. If both players give up in the same period, no one gets the object. For each period that passes before a player gives up, she incurs a cost $c \in (0, 1)$. If player 1 plans to give up in period t and player 2 plans to give up in period s , with $s > t$, then player 2 gets the object in period $t + 1$, incurring the cost $(t + 1)c$, and player 1 incurs the cost tc .

To be more specific, we can model this situation with the following payoffs:

$$u_i(t_i, t_j) = \begin{cases} -t_i c & \text{if } t_i \leq t_j, \\ 1 - (t_i + 1)c & \text{if } t_i > t_j. \end{cases}$$

- Does this game have pure-strategy equilibria? If so, characterize all of them.
 - Now consider mixed strategies. Find first the symmetric mixed strategy equilibrium. Investigate if the game also admits asymmetric mixed strategy equilibria and if so, find them.
- 3.4 The government is seeking to award a contract to supply a homogeneous good in a market whose demand is given by $D(p) = \alpha - p$ for $p \leq \alpha$ and $D(p) = 0$ for $p > \alpha$. Two firms, firm 1 and firm 2, compete for this contract by each submitting a bid consisting of a price per unit they would be willing to charge in this market.

The firm that posts the lowest bid wins the contract and obtains the exclusive rights to supply the entire market demand $D(p)$. The winning firm must then meet all market demand at the price it quoted.

Both firms have constant marginal costs of production, denoted by $c \in (0, \alpha)$. Each firm must bid with a single price quote.

In the case that both firms bid with the same price quote, both obtain the rights to supply the market by dividing it equally.

- Write down firm i 's payoff function, as a function of both its quoted price p_i and the price quoted by its rival p_j .
- Characterize firm i 's best response function.

- (c) Use the best response function of both firms to characterize the set of Nash equilibria.
- 3.5 Consider a similar setup as in the previous problem. Assume that each firm's preferences are represented by its *expected profit*. Assume that the marginal cost is strictly positive. Finally, assume that $(p - c)D(p)$ is now increasing in p and increases without bound as p increases without bound (for example, this is the case if the market demand $D(p)$ is perfectly inelastic).
- (a) Does this game has a pure-strategy Nash equilibrium?
- (b) Let $\hat{p}$ be any price such that $\hat{p} > c$. Show that for every $\hat{p}$, the game has a mixed strategy Nash equilibrium in which each firm uses the same mixed strategy F , with $F(\hat{p}) = 0$ and $F(p) > 0$ for $p > \hat{p}$.

4 Math and Game Theory

- 4.1 In this problem, you will prove a classic result called *the Minimax Theorem* (von Neumann, 1928), using concepts and tools that were not available when von Neumann first stated and proved this theorem. Consider a finite game $G = \langle I, \{A_i\}_{i \in I}, \{u_i\}_{i \in I} \rangle$ and consider its mixed extension $\Gamma = \langle I, \{\Sigma_i\}_{i \in I}, \{U_i\}_{i \in I} \rangle$. Define i 's maximin expected utility level $\underline{v}_i$ to be the level he can guarantee himself in game Γ :

$$\underline{v}_i := \max_{\sigma_i \in \Sigma_i} \min_{\sigma_{-i} \in \Sigma_{-i}} U_i(\sigma_i, \sigma_{-i}).$$

Define i 's *minimax* utility level $\bar{v}_i$ to be the worst expected utility level he can be forced to receive if he gets to respond to his rival's actions:

$$\bar{v}_i = \min_{\sigma_{-i} \in \Sigma_{-i}} \max_{\sigma_i \in \Sigma_i} U_i(\sigma_i, \sigma_{-i}).$$

- (a) Show that in any game Γ , $\bar{v}_i \geq \underline{v}_i$.
- (b) Give an example of a game in which the inequality above is strict.
- (c) A two-player finite game $G_0 = \langle I, \{A_1, A_2\}, \{u_1, u_2\} \rangle$ is called a *zero-sum* game if $u_1(a_1, a_2) = -u_2(a_1, a_2)$ for all $(a_1, a_2) \in A_1 \times A_2$. Prove that in any mixed strategy Nash equilibrium of the zero-sum mixed extension of G , that is, $\Gamma_0 = \langle I, \{\Sigma_1, \Sigma_2\}, \{U_1, U_2\} \rangle$, player i 's expected utility U_i° satisfies $U_i^\circ = \underline{v}_i = \bar{v}_i$. *Hint: Such an equilibrium exists by Nash's Theorem.*
- (d) Show that if $(\hat{\sigma}_1, \hat{\sigma}_2)$ and $(\tilde{\sigma}_1, \tilde{\sigma}_2)$ are both Nash equilibria of the zero-sum game Γ_0 , then so are $(\hat{\sigma}_1, \tilde{\sigma}_2)$ and $(\tilde{\sigma}_1, \hat{\sigma}_2)$.